

Mandatory Assignment 1

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1 Exercise 1

We download monthly data for the S&P 500 index ($\hat{GSPC}$) and the 13-week Treasury Bill rate ($\hat{IRX}$) from Yahoo! Finance for the period January 1, 2000 to February 28, 2026. The $\hat{IRX}$ series is an annualized yield in percent, so it is converted into a monthly risk-free rate as:

$$r_{f,t} = (1 + IRX_t/100)^{1/12} - 1.$$

Monthly excess market returns are then computed as: $\tilde{r}_{m,t} = r_{m,t} - r_{f,t}$, where $r_{m,t}$ denotes the monthly return on the S&P 500 index. We use the estimated monthly mean excess return μ_m and monthly variance σ_m^2 to calibrate the synthetic asset universe. Table 1 reports the estimated monthly market moments and their annualized counterparts. The estimated monthly mean excess return is approximately 0.45%, corresponding to an annualized mean excess return of approximately 5.45%. The annualized variance is approximately 0.0228.

Table 1: Estimated monthly and annualised market parameters.

Parameter	Monthly	Annualised
Mean excess return	0.0045	0.0545
Variance	0.0019	0.0228

Note: Based on monthly S&P 500 excess returns ($\hat{GSPC}$) from January 2000 to February 2026. The risk-free rate is $\hat{IRX}$, converted to a monthly rate.

Source: Yahoo! Finance.

2 Exercise 2

Next, we construct a universe of $N = 50$ synthetic assets following the single-factor model

$$\tilde{r}_{i,t} = \beta_i \tilde{r}_{m,t} + \varepsilon_{i,t}.$$

The factor loadings β_i and idiosyncratic volatilities $\sigma_{\varepsilon,i}$ are drawn from the distributions specified in the assignment. Using the estimated market moments from Exercise 1, we construct the true

mean vector μ and covariance matrix Σ , which are treated as the true parameters throughout the remainder of the assignment.

3 Exercise 3

We next implement functions to compute the tangency portfolio and the mean-variance efficient frontier for a given mean vector μ and covariance matrix Σ . The tangency portfolio maximizes the Sharpe ratio, while the efficient frontier contains the portfolios with minimum variance for a given target return.

4 Exercise 4

Using the true parameter values μ and Σ , we compute the mean-variance efficient frontier and the corresponding tangency portfolio. The efficient frontier traces the highest expected excess return attainable at each level of portfolio volatility under the true data-generating process. All quantities are reported in monthly units.

Figure 1 plots the true efficient frontier together with the individual synthetic assets and the tangency portfolio. The synthetic assets all lie below and to the right of the frontier, illustrating the diversification gains available from combining them. The tangency portfolio lies on the efficient frontier and maximizes the Sharpe ratio among fully invested portfolios.

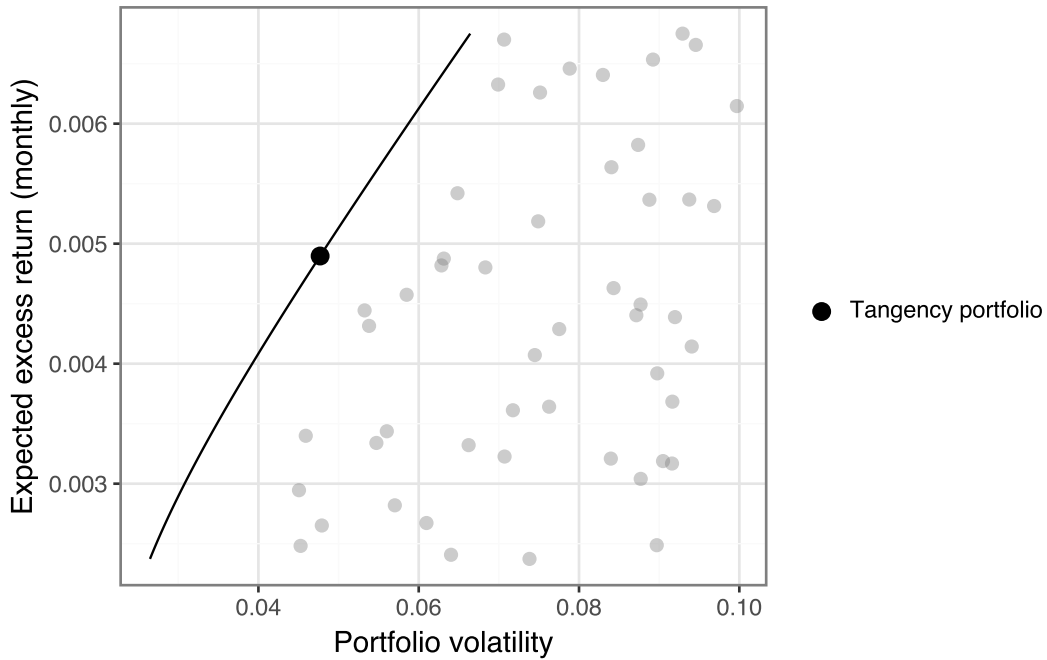


Figure 1: Mean-variance efficient frontier based on the true parameters (monthly units).

5 Exercise 5

To study estimation error, we simulate asset returns from the true data-generating process defined by μ and Σ . For each simulated sample, we estimate the sample mean vector $\hat{\mu}$ and covariance matrix $\hat{\Sigma}$. These plug-in estimates are then used in the same mean-variance frontier function as before.

This allows us to compare the efficient frontier implied by the true parameters with frontiers based on estimated sample moments. We repeat the exercise for different sample lengths T to examine how the amount of available data affects the shape and stability of the estimated frontier. Since the number of assets is fixed at $N = 50$, increasing T lowers the ratio N/T and should reduce estimation error.

We first compare the true efficient frontier with the plug-in frontier estimated from a single simulated sample with $T = 120$ monthly observations. This illustrates how sampling variation in $\hat{\mu}$ and $\hat{\Sigma}$ can change the perceived risk-return trade-off.

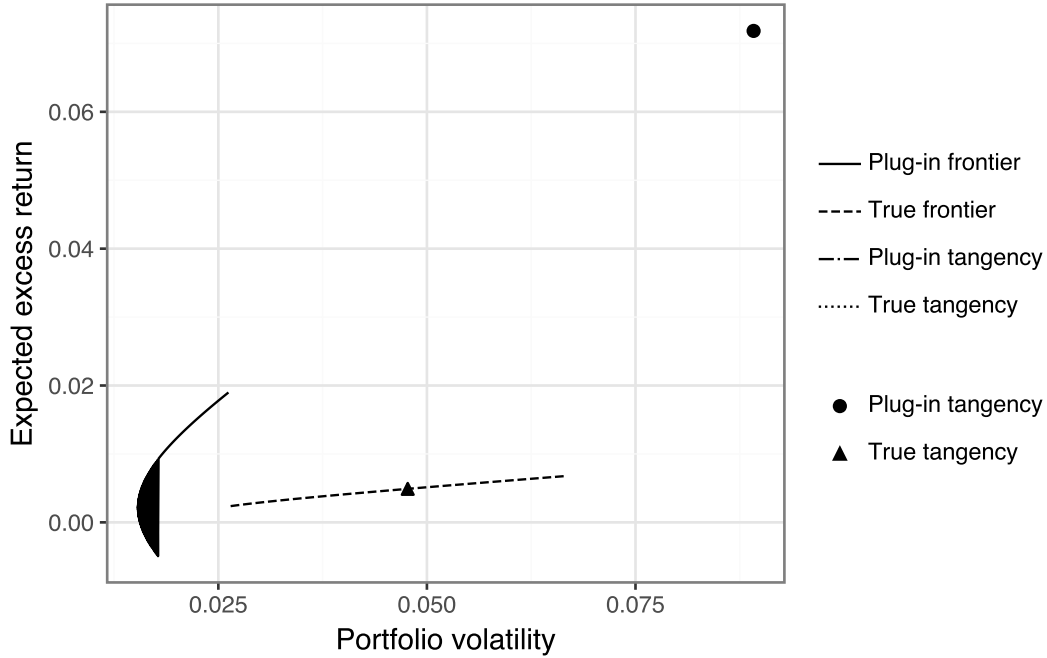


Figure 2: True efficient frontier and plug-in frontier based on one simulated sample with $T = 120$ monthly observations.

Figure 2 illustrates that a single finite sample can produce a plug-in frontier that differs substantially from the true frontier. The difference arises because the sample mean vector and covariance matrix contain sampling noise, which is then passed directly into the portfolio optimization.

To study how sample size affects estimation error, we repeat the simulation for different values of T .

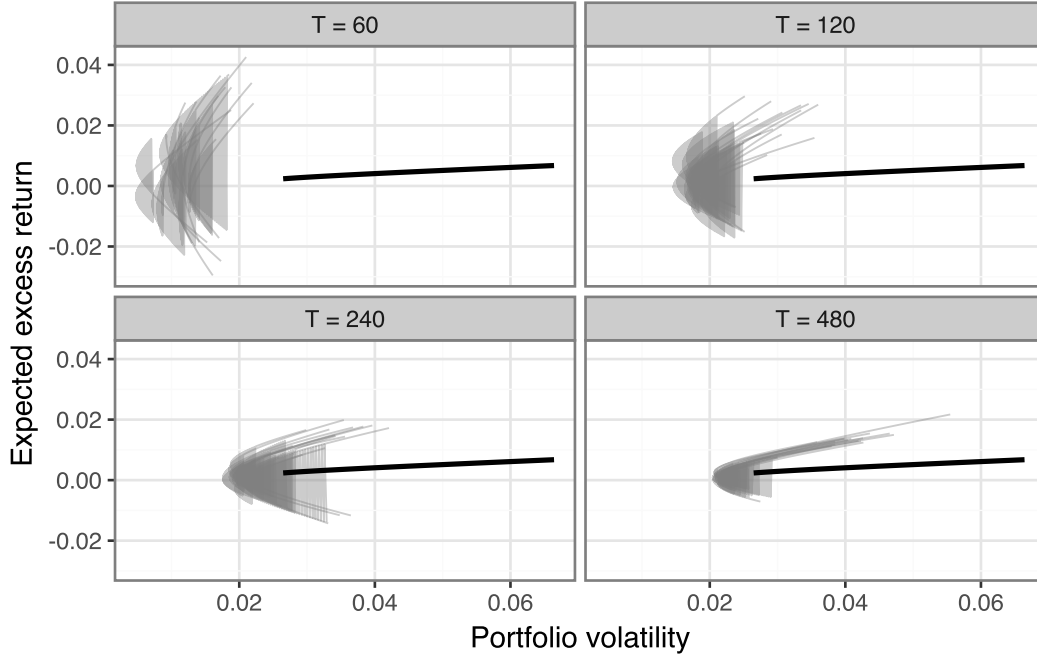


Figure 3: Plug-in efficient frontiers for different sample sizes.

Figure 3 shows that this effect is strongest when the sample size is small. For $T = 60$, the plug-in frontiers are highly dispersed, indicating substantial estimation uncertainty. As T increases, the frontiers become more concentrated around the true frontier. This illustrates that the severity of estimation error is mainly determined by the ratio N/T : with many assets relative to the number of observations, the estimated moments are noisy and the resulting efficient frontier is unstable.

6 Exercise 6

We evaluate the out-of-sample performance loss from using plug-in estimates instead of the true parameters. The true tangency portfolio is computed from the known parameters μ and Σ , while the plug-in tangency portfolio is computed from the sample estimates $\hat{\mu}$ and $\hat{\Sigma}$ obtained from simulated returns.

For excess returns, the tangency portfolio weights are:

$$w_{tan} = \frac{\Sigma^{-1}\mu}{\iota^\top \Sigma^{-1}\mu},$$

where ι denotes a vector of ones. Portfolio performance is evaluated using the true parameters. Thus, for any portfolio weights w , the true Sharpe ratio is:

$$SR(w) = \frac{w^\top \mu}{\sqrt{w^\top \Sigma w}}.$$

For each sample size T , we simulate returns from the true data-generating process, estimate $\hat{\mu}$ and $\hat{\Sigma}$, compute the plug-in tangency portfolio, and evaluate its Sharpe ratio using the true μ and Σ .

This procedure is repeated across simulations to assess how estimation error depends on the sample length.

Table 2: Average Sharpe ratios of the true and plug-in tangency portfolios across sample sizes.

	T	true_sharpe	plugin_sharpe
0	60	0.102665	0.006071
1	120	0.102665	0.009264
2	240	0.102665	0.015513
3	480	0.102665	0.024318

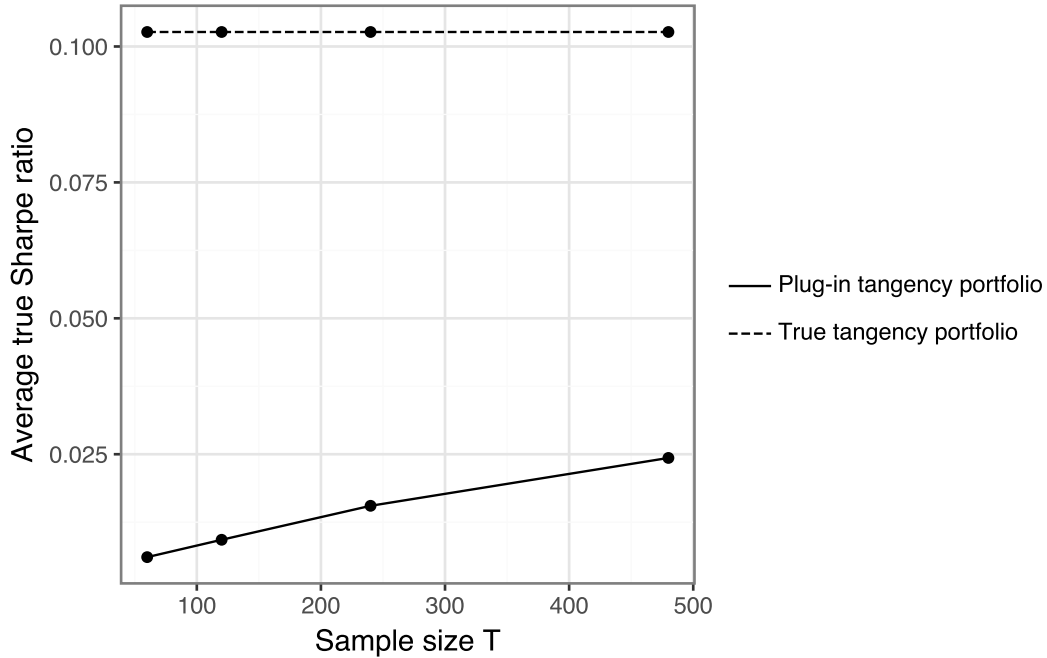


Figure 4: True and plug-in tangency portfolio performance evaluated under the true parameters.

Table 2 reports the average Sharpe ratios of the true tangency portfolio and the plug-in tangency portfolio for different sample sizes.

The results show that the plug-in tangency portfolio performs substantially worse than the true tangency portfolio when T is small. As T increases, the plug-in Sharpe ratio improves because the estimates $\hat{\mu}$ and $\hat{\Sigma}$ become more accurate.

Figure 4 illustrates the same pattern. The performance of the plug-in tangency portfolio increases with the sample size, but remains below the performance of the true tangency portfolio.

7 Exercise 7

To study the robustness of different portfolio construction methods, we compare the plug-in tangency portfolio with several alternative strategies.

The first benchmark is the equally weighted portfolio, defined as

$$w_{EW} = \frac{1}{N}\mathbf{1}.$$

This portfolio assigns equal weights to all assets and does not depend on estimated parameters.

The second strategy is the global minimum variance portfolio, which minimizes portfolio variance subject to the weights summing to one. The weights are given by

$$w_{GMV} = \frac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}^\top \Sigma^{-1}\mathbf{1}}.$$

This strategy depends only on the covariance matrix and does not require estimates of expected returns.

The third strategy is the inverse-volatility portfolio, where assets with higher volatility receive lower weights. The weights are defined as

$$w_i = \frac{1/\sigma_i}{\sum_{j=1}^N 1/\sigma_j}.$$

Table 3: Average true Sharpe ratios across portfolio strategies and sample sizes.

strategy T	Equally weighted	Global minimum variance	Inverse volatility	Plug-in tangency
60	0.1018	0.0184	0.102	0.0057
120	0.1018	0.0372	0.102	0.0086
240	0.1018	0.0433	0.102	0.0129
480	0.1018	0.0450	0.102	0.0222

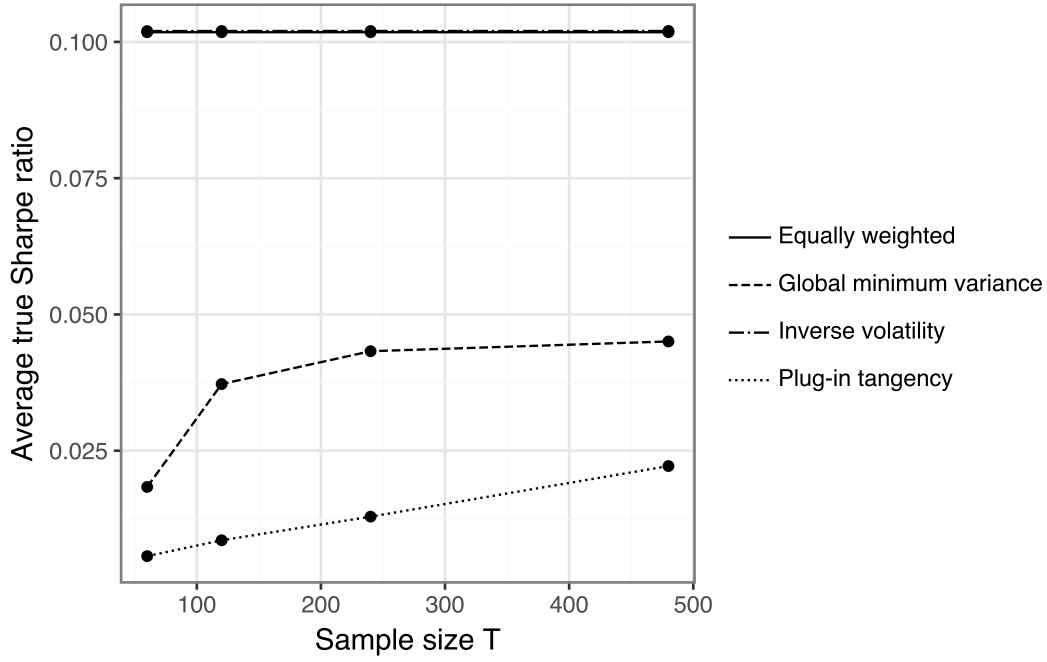


Figure 5: Average true Sharpe ratio across portfolio strategies.

Figure 5 compares the average Sharpe ratios of the different strategies across sample sizes.

The results show that the plug-in tangency portfolio is the most sensitive to estimation error. In contrast, the equally weighted and inverse-volatility portfolios perform more consistently across different sample sizes because they rely less on estimated parameters. As the sample size increases, the performance of the plug-in tangency portfolio improves and gradually approaches the true optimum.